Lecture 2

# Universal properties, adjunctions, semisimple categories

**Objectives.** *Every textbook builds $V \oplus W$ and $V \otimes W$ in its own way, with its own bases, and every textbook gets the same answer. This lecture says why: both are characterised by a universal property, which determines them up to a unique isomorphism compatible with the universal maps. The same idea gives the adjunction between $\otimes$ and* Hom*, which Lecture 4 turns into bending wires. It then explains the shape of* $\mathrm{Rep}(G)$*: a semisimple category is equivalent to $n$ copies of* $\mathrm{Vect}_{\mathrm{fd}}$*, one for each simple object.*

By the end of this lecture you will be able to:

- State the universal properties of products, coproducts and the tensor product, and prove that a representing object is unique up to a unique isomorphism compatible with the universal maps.
- Prove $\mathrm{Hom}(U \otimes V, W) \cong \mathrm{Hom}(U, \mathrm{Hom}(V, W))$, naturally, and read it as an adjunction with its unit and counit.
- Prove that a semisimple category with simple objects $X_1, \dots, X_n$ is equivalent to $\mathrm{Vect}_{\mathrm{fd}}^n$, and compute its Hom spaces from multiplicities.
- Decompose the permutation representation of $S_3$, and show that a Jordan block in $\mathrm{Rep}(\mathbb{Z})$ is not a direct sum of simple objects.

## 2.1 Universal properties

A universal property describes an object by the morphisms into it, or out of it, rather than by its elements. The simplest kind asks for exactly one.

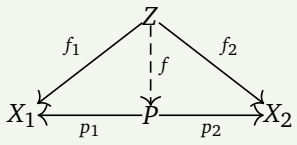
**Definition 2.1** (Terminal and initial objects)**.** An object $T$ is *terminal* if every object $X$ has exactly one morphism $X \to T$. It is *initial* if every $X$ has exactly one morphism $T \to X$.

In Set a one-point set is terminal and the empty set is initial. In Vect the zero space is both.

**Lemma 2.2.** *Any two terminal objects are isomorphic by a unique isomorphism, and so are any two initial objects.*

*Proof.* Let $T$ and $T'$ be terminal. There is exactly one morphism $a \colon T \to T'$ and one $b \colon T' \to T$. Then $b \circ a$ and $\mathrm{id}_T$ are both morphisms $T \to T$, and there is only one, so $b \circ a = \mathrm{id}_T$; likewise $a \circ b = \mathrm{id}_{T'}$. The initial case is the same with the arrows reversed. ∎

**Definition 2.3** (Product, coproduct)**.** A *product* of $X_1$ and $X_2$ is an object $P$ with morphisms $p_k\colon P \to X_k$ such that for every object $Z$ and morphisms $f_k\colon Z \to X_k$ there is exactly one $f\colon Z \to P$ with $p_k \circ f = f_k$ for $k = 1, 2$:




A *coproduct* is the same with every arrow reversed: morphisms $i_k\colon X_k \to C$ such that every pair $g_k\colon X_k \to Z$ factors as $g \circ i_k$ through exactly one $g\colon C \to Z$.

The proof of Lemma 2.2 applies word for word: a product is unique up to a unique isomorphism compatible with the projections. So "the" product is well defined, whatever construction a textbook uses. In Set the product is $X_1 \times X_2$ and the coproduct the disjoint union, and the two differ: for sets of 2 and 3 elements they have 6 and 5 elements. In Vect they coincide.

**Definition 2.4** (Biproduct)**.** In a category whose Hom sets are vector spaces, composed bilinearly, a *biproduct* of $X_1$ and $X_2$ is an object $X_1 \oplus X_2$ with morphisms $i_k\colon X_k \to X_1 \oplus X_2$ and $p_k\colon X_1 \oplus X_2 \to X_k$ such that

$$p_k \circ i_l = \delta_{kl}\ \mathrm{id}_{X_k},$$
$$i_1 \circ p_1 + i_2 \circ p_2 = \mathrm{id}\,.$$

The direct sum of vector spaces, with its inclusions and projections, is a biproduct, and a biproduct is both a product and a coproduct: given $f_k\colon Z \to X_k$, the map $f = i_1 \circ f_1 + i_2 \circ f_2$ has $p_k \circ f = f_k$, and any $f$ with this property equals $(i_1 p_1 + i_2 p_2) \circ f$, which is the same map. The coproduct is the mirror argument. This holds for finitely many summands only: the sequence $(1, 1, 1, \dots)$ lies in the product of countably many copies of $\mathbb{C}$ but not in their direct sum, whose elements have finitely many non-zero entries.

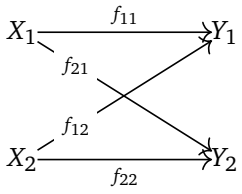
A biproduct turns morphisms into matrices. A morphism $f\colon X_1 \oplus X_2 \to Y_1 \oplus Y_2$ is determined by its four components $f_{jk} = p_j \circ f \circ i_k\colon X_k \to Y_j$, since

$$f = \sum_{j,k} i_j \circ f_{jk} \circ p_k, \tag{2.1}$$

and composites multiply like matrices,

$$(g \circ f)_{lk} = \sum_j g_{lj} \circ f_{jk}.$$

Drawn as arrows between the summands, $f$ is the picture




and composing two such pictures sums over the paths through the middle column. We use this calculus of matrices of morphisms in § 2.4.

## 2.2 Representables; Yoneda (statement and use)

The tensor product is not a product: *Linear Algebra* (Prop. 12.5) characterises it by its maps out. For every bilinear map $B\colon U\times V\to W$ there is exactly one linear map $\widetilde{B}\colon U\otimes V\to W$ with

$$\widetilde{B}(u\otimes v)=B(u,v).$$

In other words, linear maps out of $U\otimes V$ are the same as bilinear maps, naturally in $W$. Universal properties of this shape have a name.

**Definition 2.5** (Representable functor)**.** A functor $\Phi\colon C\to\mathrm{Set}$ is *represented* by an object $X$ if there is a natural isomorphism

$$\mathrm{Hom}(X,-)\cong\Phi.$$

Here $\mathrm{Hom}(X,-)$ sends $Y$ to $\mathrm{Hom}(X,Y)$ and $g\colon Y\to Y'$ to $f\mapsto g\circ f$.

So $U\otimes V$ represents $W\mapsto\{\text{bilinear maps } U\times V\to W\}$, and the coproduct $X_1\oplus X_2$ represents $Z\mapsto\mathrm{Hom}(X_1,Z)\times\mathrm{Hom}(X_2,Z)$. (The product represents a functor of the other variance, $Z\mapsto\mathrm{Hom}(Z,X_1)\times\mathrm{Hom}(Z,X_2)$, by $\mathrm{Hom}(-,P)$; everything below holds for it with the arrows reversed.)

**Theorem 2.6** (Yoneda · *cited*)**.** *For every functor $\Phi\colon C\to\mathrm{Set}$ and object $X$, the natural transformations $\mathrm{Hom}(X,-)\Rightarrow\Phi$ correspond bijectively to the elements of $\Phi(X)$, by $\xi\mapsto\xi_X(\mathrm{id}_X)$ (Leinster, Thm. 4.2.1 and Cor. 4.3.3).*

We need only one consequence, and we prove it directly.

**Proposition 2.7** (A representing object is unique)**.** *Let $\xi\colon\mathrm{Hom}(X,-)\Rightarrow\mathrm{Hom}(X',-)$ be a natural isomorphism. Then $\varphi=\xi_X(\mathrm{id}_X)\colon X'\to X$ is an isomorphism, and it is the only morphism with*

$$\xi_Y(f)=f\circ\varphi$$

*for every $Y$ and every $f\colon X\to Y$.*

*Proof.* Naturality for $f\colon X\to Y$, applied to $\mathrm{id}_X$, gives

$$\begin{aligned}\xi_Y(f)&=\xi_Y(f\circ\mathrm{id}_X)\\&=f\circ\xi_X(\mathrm{id}_X)=f\circ\varphi.\end{aligned}$$

Any $\varphi'$ with this property has $\varphi'=\xi_X(\mathrm{id}_X)=\varphi$. Likewise $\psi=\xi_{X'}^{-1}(\mathrm{id}_{X'})\colon X\to X'$ has $\xi_Y^{-1}(g)=g\circ\psi$. Hence

$$\begin{aligned}\mathrm{id}_X&=\xi_X^{-1}(\varphi)=\varphi\circ\psi,\\\mathrm{id}_{X'}&=\xi_{X'}(\psi)=\psi\circ\varphi.\end{aligned}$$

∎

If $X$ and $X'$ both represent $\Phi$, composing one natural isomorphism with the inverse of the other gives such an $\xi$. So two constructions of $U\otimes V$, or of $X_1\oplus X_2$, are isomorphic by exactly one isomorphism that respects the universal property. That is the precise sense in which every textbook's tensor product is "the same".

## 2.3 Adjunctions; ⊗–Hom

A universal property relates morphisms of one kind to morphisms of another. When this happens for all objects at once, the two functors involved are adjoint.

**Definition 2.8** (Adjunction)**.** Functors $F\colon C \to \mathcal{D}$ and $G\colon \mathcal{D} \to C$ are *adjoint*, $F \dashv G$, if there are bijections

$$\begin{aligned}&\operatorname{Hom}_{\mathcal{D}}(F(A), B)\\ &\quad\cong \operatorname{Hom}_{C}(A, G(B)),\end{aligned}$$

natural in $A$ and in $B$. The image of $\mathrm{id}_{F(A)}$ is the *unit* $A \to G(F(A))$; the preimage of $\mathrm{id}_{G(B)}$ is the *counit* $F(G(B)) \to B$.

Fix a finite-dimensional space $V$, and write $\operatorname{Hom}(V, W)$ for the space of linear maps, itself an object of $\mathrm{Vect}_{\mathrm{fd}}$.

**Theorem 2.9** (⊗ and Hom are adjoint)**.** *The maps*

$$\begin{aligned}\Theta\colon\ &\operatorname{Hom}(U \otimes V, W)\\ &\to \operatorname{Hom}(U, \operatorname{Hom}(V, W)),\end{aligned}$$

$$\Theta(f)(u)(v) = f(u \otimes v),$$

*are bijections, natural in $U$ and in $W$. So $- \otimes V \dashv \operatorname{Hom}(V, -)$ on $\mathrm{Vect}_{\mathrm{fd}}$.*

*Proof.* $\Theta(f)(u)$ is linear in $v$, and $\Theta(f)$ is linear in $u$, because $f$ is linear and $\otimes$ is bilinear. Conversely, for $g\colon U \to \operatorname{Hom}(V, W)$ the map $(u, v) \mapsto g(u)(v)$ is bilinear, so by the universal property it is $f(u \otimes v)$ for exactly one linear $f$, and $\Theta(f) = g$. Since the $u \otimes v$ span $U \otimes V$, $f$ is determined by $\Theta(f)$, so $\Theta$ is a bijection.

Naturality in $U$: for $a\colon U' \to U$, put $f' = f \circ (a \otimes \mathrm{id}_V)$; then

$$\begin{aligned}\Theta(f')(u')(v) &= f(a(u') \otimes v)\\ &= \big(\Theta(f) \circ a\big)(u')(v).\end{aligned}$$

Naturality in $W$: for $b\colon W \to W'$, $\Theta(b \circ f)(u) = b \circ \Theta(f)(u)$, both sending $v$ to $b(f(u \otimes v))$. ∎

For a concrete instance, take $U = V = \mathbb{C}^2$ and $W = \mathbb{C}$. In the ordered tensor basis $e_1 \otimes e_1, e_1 \otimes e_2, e_2 \otimes e_1, e_2 \otimes e_2$, let the linear functional $f$ have matrix

$$f = \begin{pmatrix} 1 & 2 & 3 & -1 \end{pmatrix}.$$

Currying changes the organisation of the inputs, without changing the value:

$$\Theta(f)(u) = \begin{pmatrix} u_1 + 3u_2 & 2u_1 - u_2 \end{pmatrix}.$$

Thus $\Theta(f)(u)$ is a linear functional on $V$, rather than a number. It becomes a number only after evaluation at $v$. For $u = (2, -1)$ and $v = (1, 3)$, both evaluations give

$$\Theta(f)(u)(v) = \begin{pmatrix} -1 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = 14 = f(u \otimes v).$$

The same pattern in Set is currying: a function $X \times B \to C$ of two variables is a function of the first with values in functions of the second, so $- \times B \dashv (-)^B$; for finite sets both sides have $|C|^{|X||B|}$ elements.

The unit and the counit of Theorem 2.9 are maps every physicist has used: the unit $U \to \mathrm{Hom}(V, U \otimes V)$ and the counit $\mathrm{Hom}(V,W) \otimes V \to W$,

$$u \mapsto (v \mapsto u \otimes v),$$
$$g \otimes v \mapsto g(v).$$

They satisfy the two *triangle identities* that every adjunction satisfies (Leinster Lemma 2.2.2): inserting a vector and evaluating gives back what one started with,

$$u \otimes v \mapsto (v' \mapsto u \otimes v') \otimes v \mapsto u \otimes v,$$
$$g \mapsto (v \mapsto g \otimes v) \mapsto (v \mapsto g(v)) = g.$$

The counit is evaluation. For $W = \mathbb{C}$ it is the pairing $V^* \otimes V \to \mathbb{C}$, $\varphi \otimes v \mapsto \varphi(v)$, which Lecture 4 calls $\mathrm{ev}_V$ and draws as a cap; there the adjunction becomes the bending of a wire.

## 2.4 Linear and semisimple categories; Schur

From now on the categories have the structure of $\mathrm{Vect}_{\mathrm{fd}}$ and $\mathrm{Rep}(G)$: morphisms can be added and scaled.

**Definition 2.10** (Linear category)**.** A $\mathbb{C}$-*linear* category has a complex vector space structure on each $\mathrm{Hom}(X,Y)$, with composition bilinear. It is *additive* if it has a zero object and a biproduct of any two objects.

$\mathrm{Vect}_{\mathrm{fd}}$, $\mathrm{Rep}(G)$ and $\mathrm{Mat}_{\mathbb{C}}$ are $\mathbb{C}$-linear and additive. So is $\mathrm{Vect}_{\mathrm{fd}}^n$, whose objects are $n$-tuples $(V_1, \dots, V_n)$ of finite-dimensional spaces and whose morphisms are $n$-tuples of linear maps, composed and added slot by slot.

**Definition 2.11** (Semisimple category)**.** A $\mathbb{C}$-linear additive category with finite-dimensional Hom spaces is *semisimple*, with *simple objects* $X_1, \dots, X_n$, all nonzero, if

(S1) $\mathrm{Hom}(X_i, X_j) = 0$ for $i \neq j$, and $\mathrm{Hom}(X_i, X_i) = \mathbb{C}\,\mathrm{id}$;

(S2) every object is isomorphic to a finite direct sum of copies of the $X_i$.

For $\mathrm{Rep}(G)$ with $G$ finite, (S1) is Schur's lemma and (S2) is Maschke's theorem (*Symmetries* Thms. 4.12–4.13 and 4.8), and there are as many irreducible representations as conjugacy classes (*Symmetries* Thm. 5.12). $\mathrm{Rep}(\mathrm{SU}(2))$, with continuous representations, satisfies both conditions for its irreducible representations, one for each spin $j = 0, \frac{1}{2}, 1, \dots$ (*Symmetries* Cor. 12.5, Thm. 14.3 and Schur, Thms. 4.12–4.13), but there are infinitely many of them: it is semisimple only in the wider sense that allows an infinite list. We take (S1) and (S2) as the definition, since they are all that is used.

**Lemma 2.12** (Multiplicities)**.** *If $X \cong \bigoplus_i X_i^{\oplus m_i}$, then $m_i = \dim \mathrm{Hom}(X_i, X)$. So the multiplicities do not depend on the decomposition.*

*Proof.* By (2.1), a morphism into a biproduct is a column of morphisms into its summands, so

$$\mathrm{Hom}\Big(X_i, \bigoplus_j X_j^{\oplus m_j}\Big)$$
$$\cong \bigoplus_j \mathrm{Hom}(X_i, X_j)^{\oplus m_j}.$$

By (S1) only $j = i$ contributes, and it gives $\mathbb{C}^{m_i}$. ∎

**Theorem 2.13** (Schur, categorically)**.** *Let $C$ be semisimple with simple objects $X_1, \dots, X_n$. The functor*

$$\begin{aligned} H &\colon C \to \mathrm{Vect}_{\mathrm{fd}}^{n}, \\ H(X) &= \big(H_i(X)\big)_{i=1}^{n}, \\ H_i(X) &= \mathrm{Hom}(X_i, X), \end{aligned}$$

*with $H(f)$ acting by $g \mapsto f \circ g$ in each slot, is an equivalence. In particular*

$$\begin{aligned} &\mathrm{Hom}(X, Y) \\ &\quad\cong \bigoplus_i \mathrm{Hom}\big(H_i(X), H_i(Y)\big) \\ &\quad\cong \bigoplus_i H_i(X)^* \otimes H_i(Y), \end{aligned}$$

*and $\dim \mathrm{Hom}(X,Y) = \sum_i m_i(X)\, m_i(Y)$, with $m_i$ the multiplicities.*

*Proof. $H$ is fully faithful.* Choose decompositions $X \cong \bigoplus_i X_i^{\oplus m_i}$ and $Y \cong \bigoplus_i X_i^{\oplus k_i}$ (S2). By (2.1) a morphism $f\colon X \to Y$ is a matrix of morphisms between copies of the $X_i$; by (S1) the only non-zero blocks join copies of the same $X_i$, and each entry is a scalar. So $f$ is a family of complex $k_i \times m_i$ matrices $A_i$, one for each $i$. In the bases of $\mathrm{Hom}(X_i, X) \cong \mathbb{C}^{m_i}$ and $\mathrm{Hom}(X_i, Y) \cong \mathbb{C}^{k_i}$ given by the inclusions of the copies of $X_i$, $H(f)$ acts in slot $i$ by the same matrix $A_i$. Hence $f \mapsto H(f)$ is a bijection.

*$H$ is essentially surjective.* Given $(V_1, \dots, V_n)$, the object $\bigoplus_i X_i^{\oplus \dim V_i}$ has $H$-image isomorphic to it, by Lemma 2.12.

*An inverse.* Choose a basis of every finite-dimensional space. Let $G(V) = \bigoplus_i X_i^{\oplus \dim V_i}$, and let $G$ send a tuple of linear maps to the block matrix of their matrices. This is a functor because matrices multiply.

In $H(G(V))$, slot $i$ has the basis of inclusions of the copies of $X_i$, and sending it to the chosen basis of $V_i$ is a natural isomorphism $H \circ G \Rightarrow \mathrm{id}$, since $G(g)$ is the matrix of $g$.

In the other direction, the chosen basis vectors $b_{ij}$ of $H_i(X)$ are morphisms $X_i \to X$; together they form a morphism $\zeta_X\colon G(H(X)) \to X$ whose restriction to the $j$-th copy of $X_i$ is $b_{ij}$.

For invertibility, observe that $H(\zeta_X)$ sends the inclusion basis to the basis $(b_{ij})$, so it is invertible; fullness gives an $h$ with $H(h) = H(\zeta_X)^{-1}$, and faithfulness gives $h \circ \zeta_X = \mathrm{id}$ and $\zeta_X \circ h = \mathrm{id}$.

For naturality, $f \circ \zeta_X$ and $\zeta_Y \circ G(H(f))$ agree on each copy of $X_i$, where both give $f \circ b_{ij}$, so $\zeta$ is a natural isomorphism $G \circ H \Rightarrow \mathrm{id}$.

The Hom formula is full faithfulness, followed by $\mathrm{Hom}(A, B) \cong B \otimes A^* \cong A^* \otimes B$ (*Linear Algebra* Prop. 12.18). Its dimension is $\sum_i k_i m_i$. ∎

**Example 2.14** (Endomorphisms with multiplicity)**.** In $\mathrm{Rep}(S_3)$ let $X = \mathbf{1} \oplus \mathbf{1} \oplus \mathrm{std}$, with $\mathbf{1}$ trivial and std the standard representation on the sum-zero plane (*Symmetries* §4.4). The theorem gives

$$\mathrm{End}(X) \cong M_2(\mathbb{C}) \times \mathbb{C},$$

of dimension $2^2 + 1^2 = 5$: any linear map between the two trivial summands is an intertwiner, and on std only scalars are. Multiplicity two makes $\mathrm{End}(X)$ non-commutative.

For the same object $X$, an intertwiner can mix the two copies of $\mathbf{1}$. It cannot send either copy into std, since $\mathrm{Hom}(\mathbf{1}, \mathrm{std}) = 0$. For example, choose endomorphisms $f$ and $g$ whose multiplicity-space maps are

$$H(f) = \left(\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, 3\right), \qquad H(g) = \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, -1\right),$$
$$H(g \circ f) = \left(\begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}, -3\right).$$

The scalar 3 means $3\,\mathrm{id}_{\mathrm{std}}$, which acts on a two-dimensional space. Thus the underlying map $f$ has a matrix of size $4 \times 4$, whereas $H(f)$ records one block of size $2 \times 2$ and one scalar. The block sizes count copies of a simple object, not its vector-space dimension. Composition multiplies the blocks separately, in the order $H(g)H(f)$.

The theorem answers the second half of the question: up to equivalence, a semisimple category is a list of vector spaces, one for each simple object, and a morphism is a list of matrices. Only the number of simple objects survives. The abelian group $\mathbb{Z}_3$ also has three irreducible representations, the characters $k \mapsto \omega^{jk}$ with $\omega = e^{2\pi i/3}$, so

$$\mathrm{Rep}(S_3) \simeq \mathrm{Rep}(\mathbb{Z}_3) \simeq \mathrm{Vect}_{\mathrm{fd}}^3$$

as linear categories, although $S_3$ is not abelian and has a two-dimensional irreducible representation. The linear category cannot tell the two groups apart. What can is the tensor product of representations, which Lecture 3 adds to the structure, and the dimensions of the underlying spaces, which an equivalence is free to forget.

## 2.5 $\mathrm{Rep}(S_3)$ and a Jordan block

**Example 2.15** ($\mathrm{Rep}(S_3)$)**.** $S_3$ has three conjugacy classes, so $\mathrm{Rep}(S_3)$ has three simple objects: the trivial representation $\mathbf{1}$, the sign representation sgn, and the two-dimensional standard representation std on the plane $x_1 + x_2 + x_3 = 0$ (*Symmetries* §4.4, with the basis $e_1 - e_2$, $e_2 - e_3$). For the permutation representation $\mathbb{C}^3$ of Lecture 1,

$$\dim \mathrm{Hom}(\mathbf{1}, \mathbb{C}^3) = 1,$$
$$\dim \mathrm{Hom}(\mathrm{sgn}, \mathbb{C}^3) = 0,$$
$$\dim \mathrm{Hom}(\mathrm{std}, \mathbb{C}^3) = 1,$$

so $\mathbb{C}^3 \cong \mathbf{1} \oplus \mathrm{std}$, and by Theorem 2.13 $\dim \mathrm{End}(\mathbb{C}^3) = 1^2 + 0^2 + 1^2 = 2$. So $\mathrm{End}(\mathbb{C}^3)$ is spanned by the identity $I$ and Lecture 1's composite $u \circ s$, the matrix of ones. With the projections $P_{\mathbf{1}} = (u \circ s)/3$ onto the line $\mathbb{C}(1,1,1)$ and $P_{\mathrm{std}} = I - P_{\mathbf{1}}$ onto the plane,

$$aI + b\,(u \circ s) = (a + 3b)\,P_{\mathbf{1}} + a\,P_{\mathrm{std}},$$

so $H$ sends it to the pair of scalars $(a + 3b, a)$, one for each simple summand. The regular representation $\mathbb{C}[S_3]$ is $\mathbf{1} \oplus \mathrm{sgn} \oplus \mathrm{std}^{\oplus 2}$, so

$$\dim \mathrm{Hom}(\mathbb{C}^3, \mathbb{C}[S_3]) = 1 \cdot 1 + 0 \cdot 1 + 1 \cdot 2 = 3.$$

The dimensions of the simple objects, 1, 1 and 2, are not seen by $H$, but they recover the dimension of every representation from its multiplicities:

$$\dim \mathbb{C}^3 = 1 \cdot 1 + 0 \cdot 1 + 1 \cdot 2 = 3,$$
$$\dim \mathbb{C}[S_3] = 1 \cdot 1 + 1 \cdot 1 + 2 \cdot 2 = 6.$$

In Lecture 4 these numbers return as quantum dimensions, computed inside the category by closing a loop.

Semisimplicity is a hypothesis, and it can fail. Maschke's theorem needs $G$ finite.

**Example 2.16** (A Jordan block)**.** Let $\mathbb{Z}$ act on $\mathbb{C}^2$ by

$$\rho(k) = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix},$$

a representation since $\rho(k+l) = \rho(k)\rho(l)$ (*Symmetries* Ex. 4.9). The line $\mathbb{C}e_1$ is invariant. An invariant complement would be a second invariant line, an eigenline of $\rho(1)$; but $\rho(1)$ has the single eigenvalue 1 with the single eigenline $\mathbb{C}e_1$. So $\mathbb{C}^2$ is not a direct sum of two nonzero representations: it is *indecomposable*. Nor does it satisfy (S1) on its own, since $\mathrm{End}(\mathbb{C}^2)$ contains $\rho(1) - I \neq 0$ besides the scalars. So $\mathbb{C}^2$ is not a direct sum of objects satisfying (S1), $\mathrm{Rep}(\mathbb{Z})$ does not satisfy (S2), and Theorem 2.13 does not apply.

The fusion categories of Lecture 5 are semisimple by definition, which excludes such examples; every count of simple objects and every multiplicity from there on rests on (S1) and (S2).

**Summary.**

- *Universal properties:* terminal objects, products, coproducts and the tensor product are determined up to a unique isomorphism compatible with the universal maps (Lemma 2.2 and Proposition 2.7); in Vect the direct sum is a biproduct, and morphisms between biproducts are matrices of morphisms, (2.1).
- *Adjunction:* $\mathrm{Hom}(U \otimes V, W) \cong \mathrm{Hom}(U, \mathrm{Hom}(V, W))$ naturally, so $- \otimes V \dashv \mathrm{Hom}(V, -)$, with evaluation as the counit (Theorem 2.9).
- *The main result:* a semisimple category with simple objects $X_1, \dots, X_n$ is equivalent to $\mathrm{Vect}_{\mathrm{fd}}^n$, and $\dim \mathrm{Hom}(X, Y) = \sum_i m_i(X) m_i(Y)$ (Theorem 2.13); $\mathrm{Rep}(S_3)$ has three simples, and $\mathbb{C}^3 \cong \mathbf{1} \oplus \mathrm{std}$.
- *The guard:* a Jordan block in $\mathrm{Rep}(\mathbb{Z})$ is indecomposable but not simple, so semisimplicity can fail (Example 2.16).
- *Next:* Lecture 3 adds the tensor product to the category, and asks what makes putting systems side by side associative.

## References

Leinster, *Basic Category Theory*: adjunctions, Def. 2.1.1, p. 41, unit and counit, §2.2, p. 51; initial and terminal objects, Def. 2.1.7 and Lemma 2.1.8, pp. 48–49; representables, Def. 4.1.3, p. 84; Yoneda, Thm. 4.2.1, p. 94, and Cor. 4.3.3, p. 100; products, Def. 5.1.1, p. 108; the direct sum as product and sum, Exs. 5.1.5 and 5.2.5, pp. 110 and 127–128. Mac Lane, *Categories for the Working Mathematician*: the tensor–hom adjunction, §IV.1, p. 80;

biproducts, §VIII.2, p. 194. Etingof, Gelaki, Nikshych and Ostrik, *Tensor Categories*: additive and linear categories, Defs. 1.2.1–1.2.2, pp. 1–2; simple and semisimple objects and Schur's lemma, Def. 1.5.1 and Lemma 1.5.2, p. 5; $\mathrm{End}(X) = k$, Prop. 1.8.4, p. 9.